

Tugboat

A cloud application to streamline big data archiving

A joint talk by

NOAA NCEI / CIRES

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and

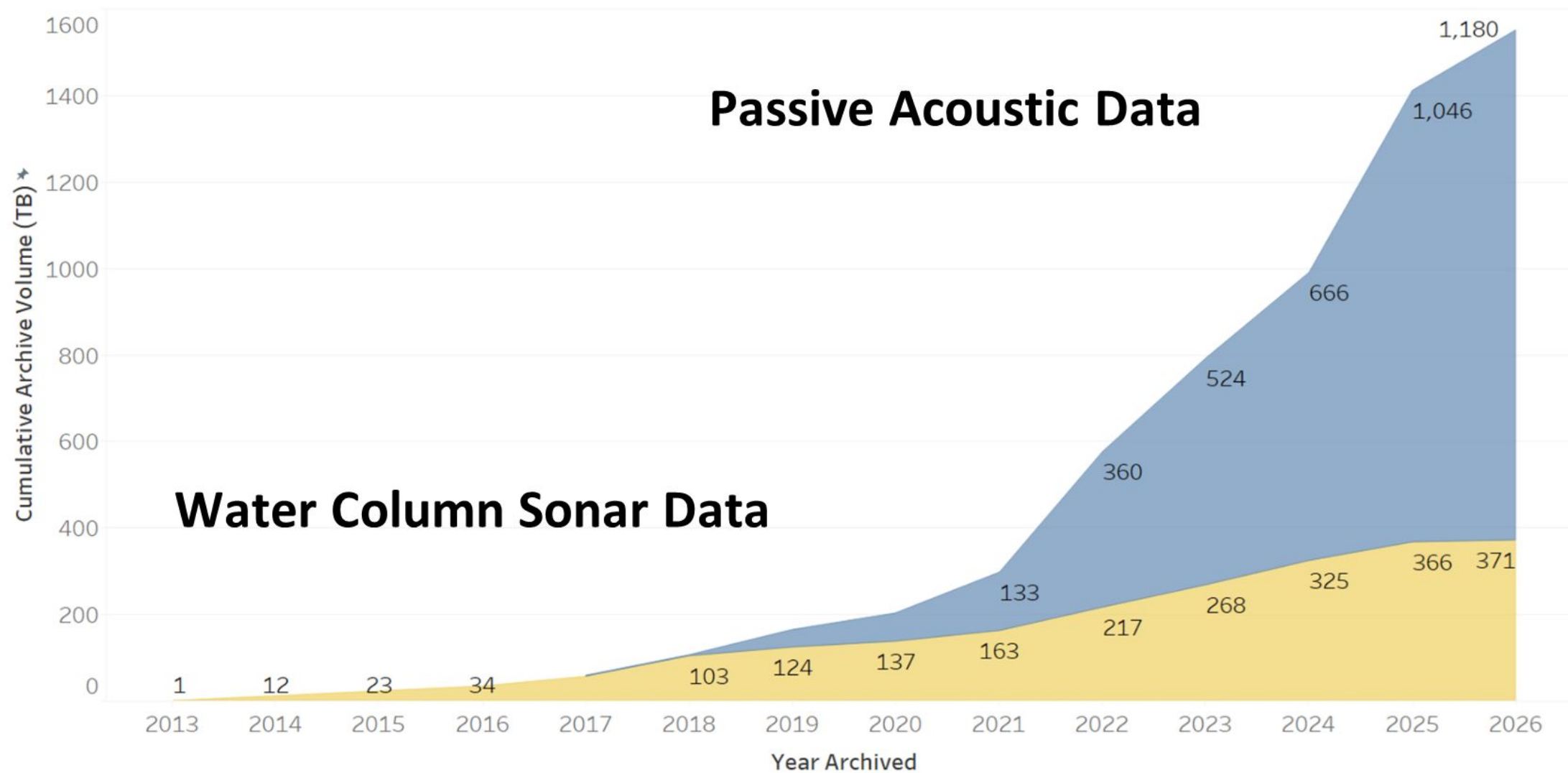
NOAA Fisheries

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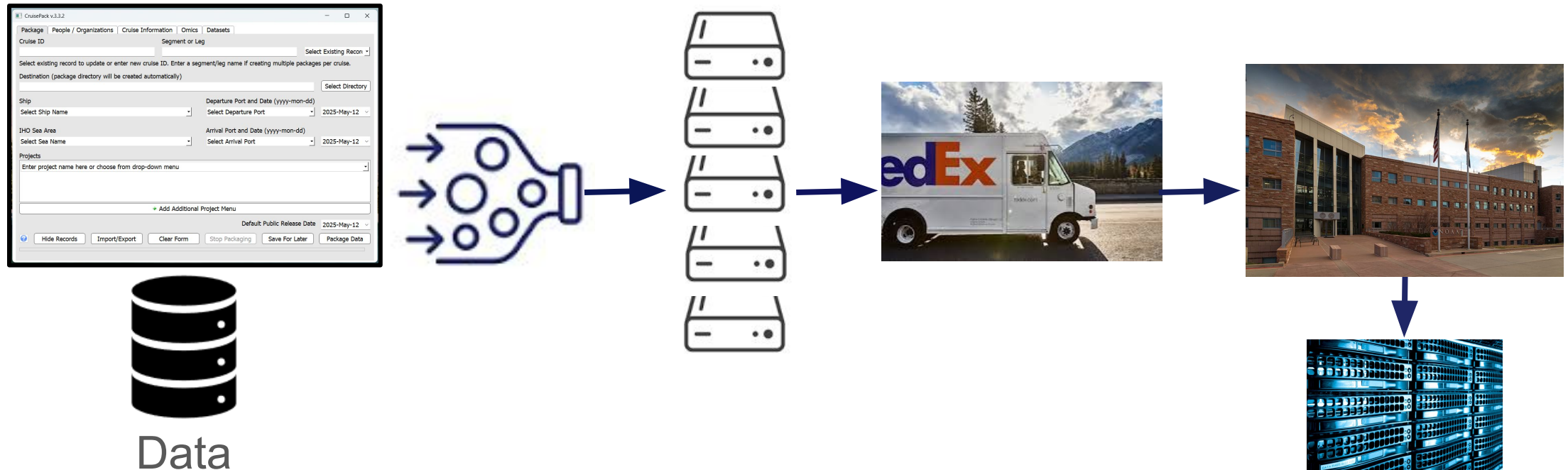


Ocean Acoustic Archives



Challenges

Traditional Pathway for the Ocean Acoustic Archive is Cumbersome Metadata



NOAA Cloud Migrations

NOAA OMAO and NOAA Fisheries are moving at an **accelerated pace to the cloud** for their data storage and processing. **NCEI, in turn, needs to adapt** its abilities to support preparation and receipt of data from cloud-based environments, especially for **big data like active and passive acoustics**

Tugboat

Tugboat is a new job orchestration and metadata management service within NCEI.

- Guides and manages the complex movement of data through validation, ingestion, and archival systems
- Vastly expedites the preparation and submission of NOAA acoustic datasets for archive at NCEI



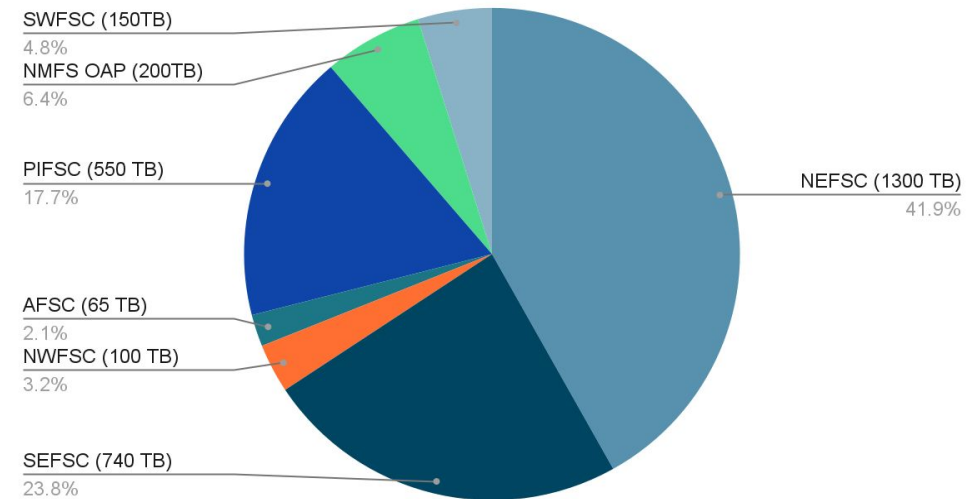
Passive Acoustic Monitoring

PAM presents an opportunity for long-term, global synthesis of ocean trends

But, realizing this potential requires:

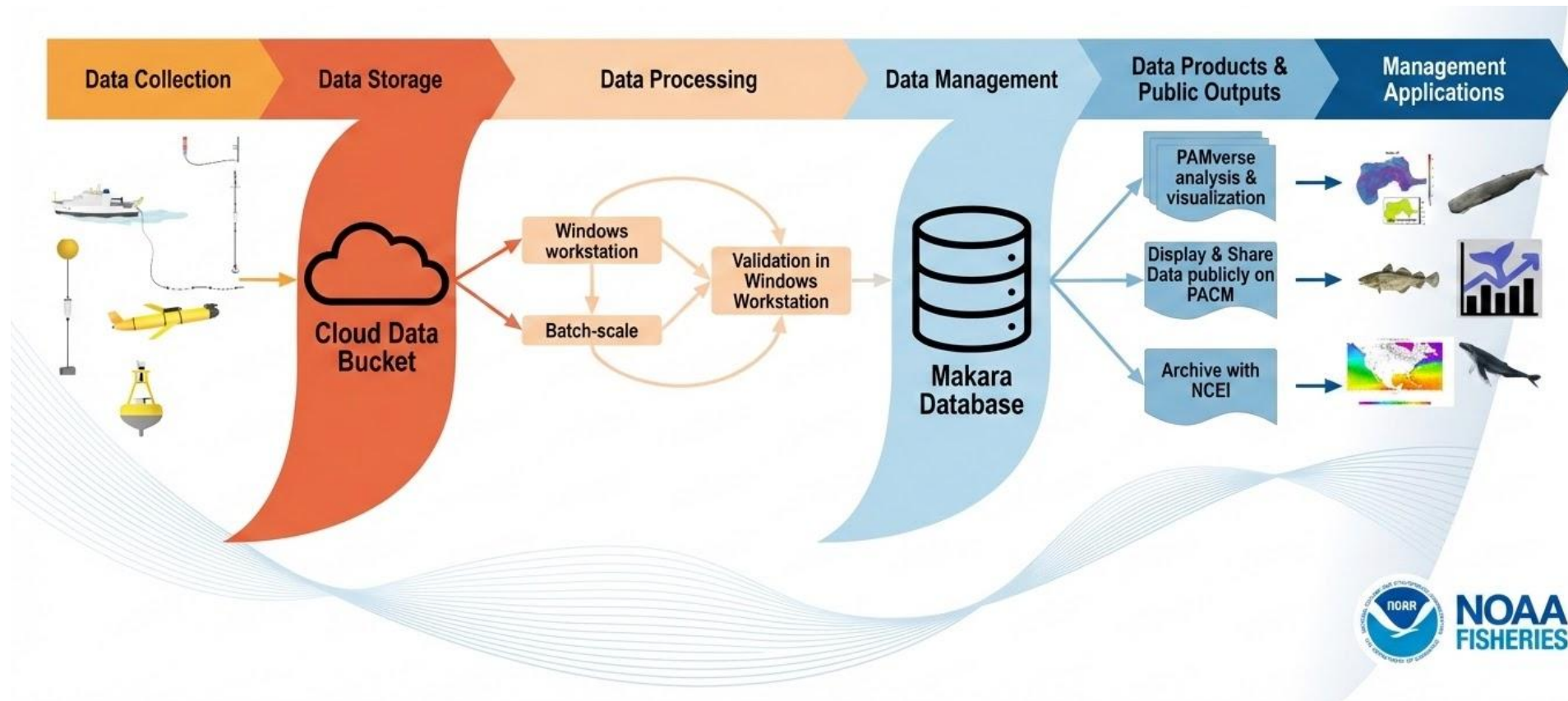
- Standardized metadata
- Streamlined data workflows
- Collaboration across NOAA, state partners, universities, NGOs, and industry
- Standardized, comparable data products to support science and management
- Data availability through archiving

NOAA Fisheries PAM data: >3 Petabytes

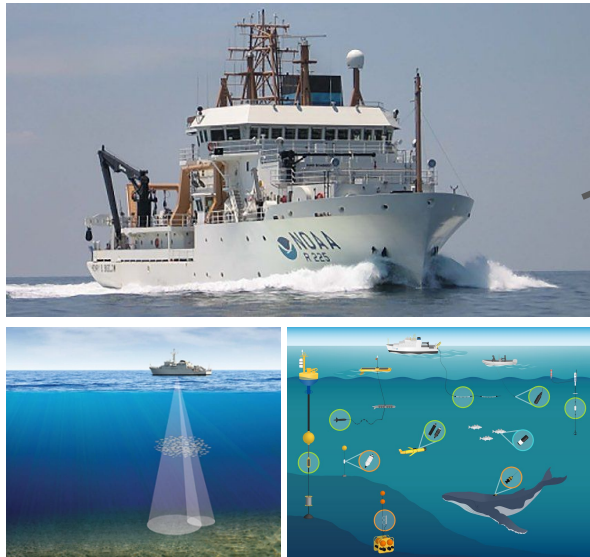


NOAA Fisheries Passive Acoustic Monitoring Strategic Initiative (PAM SI)

Creating standardized, cloud-based, and open source workflows for PAM data



Data Acquisition



- Collection of both:
- Water column sonar data
 - Passive acoustic data

Cloud Storage *Metadata & Data*

Fisheries
GCP bucket



OMAO
Azure bucket



Strategic Initiatives:

- Active Acoustic (AA-SI)
- Passive Acoustic Monitoring (PAM SI)

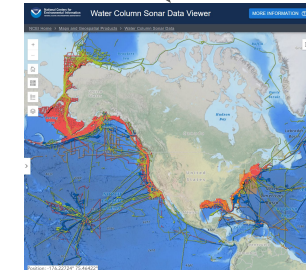


NCEI Archive

Ocean Acoustic
Archive
Pipelines

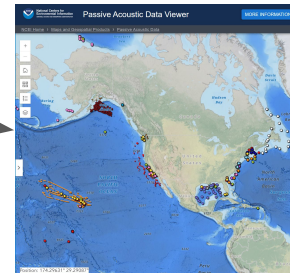
NCEI-CO
Archive

Oracle
Database



Discovery & Access

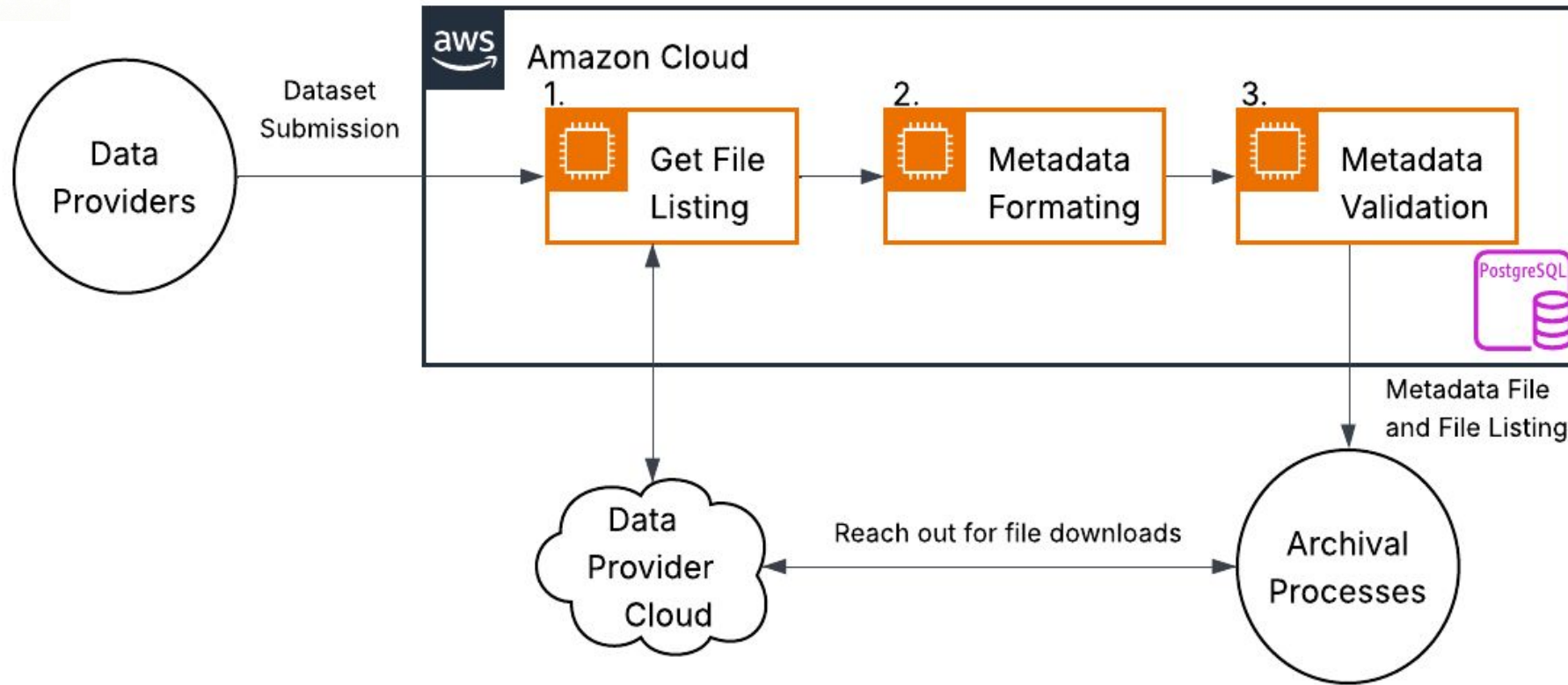
NODD
AWS & GCP





How Tugboat Works

NCEI perspective



Important Considerations for Using Tugboat

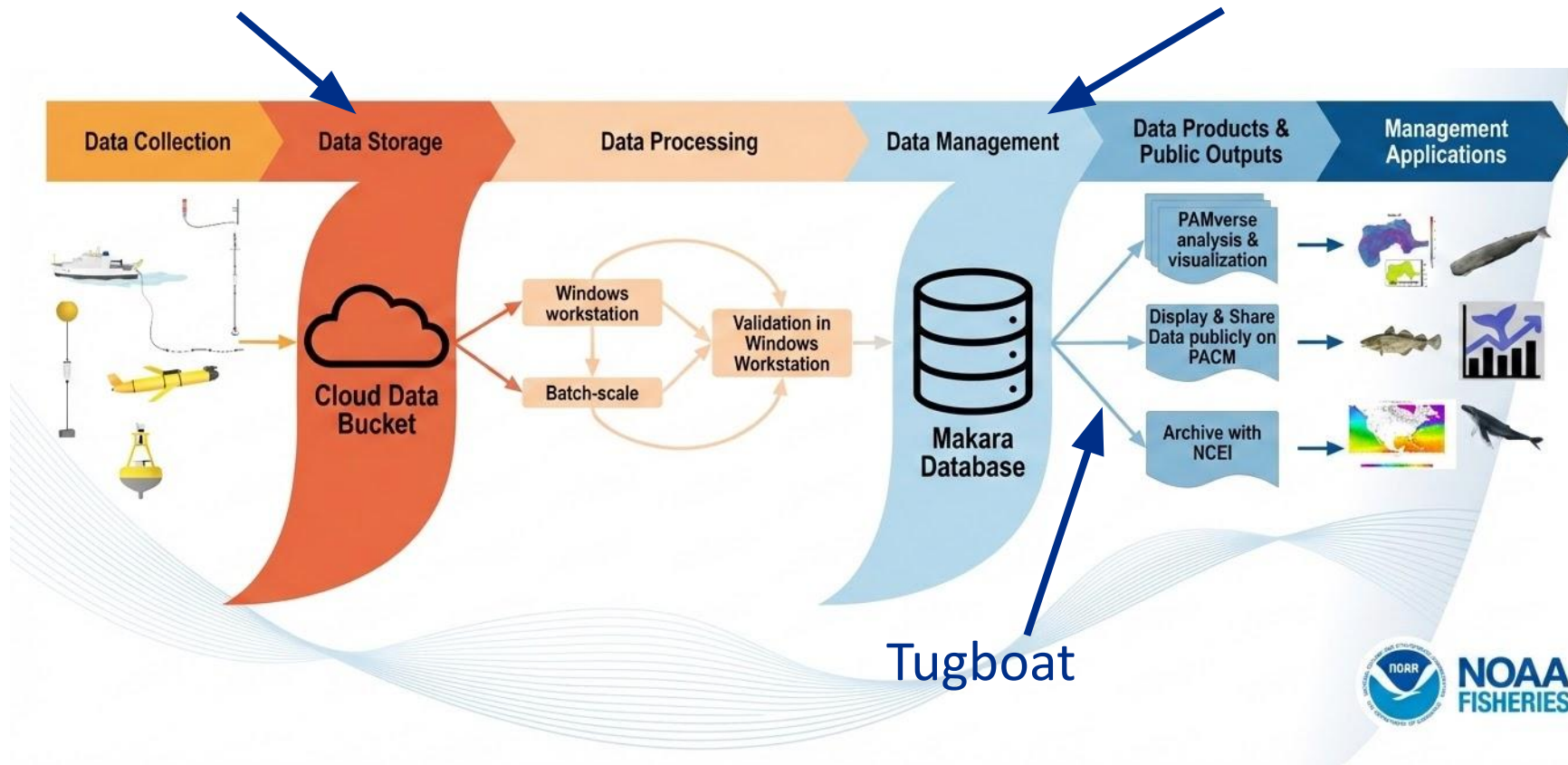
- Metadata
 - Where is it coming from?
 - Who is responsible for accuracy and completeness?
- Data
 - Data must be well organized
 - Data files, calibration files, and other ancillary files must all live in separate, non-nested, cloud locations
- Submission to NCEI
 - Who decides when a dataset and metadata are ready for submission?
 - How is information assembled and submitted to Tugboat?

How Tugboat Works

PAM-SI Perspective

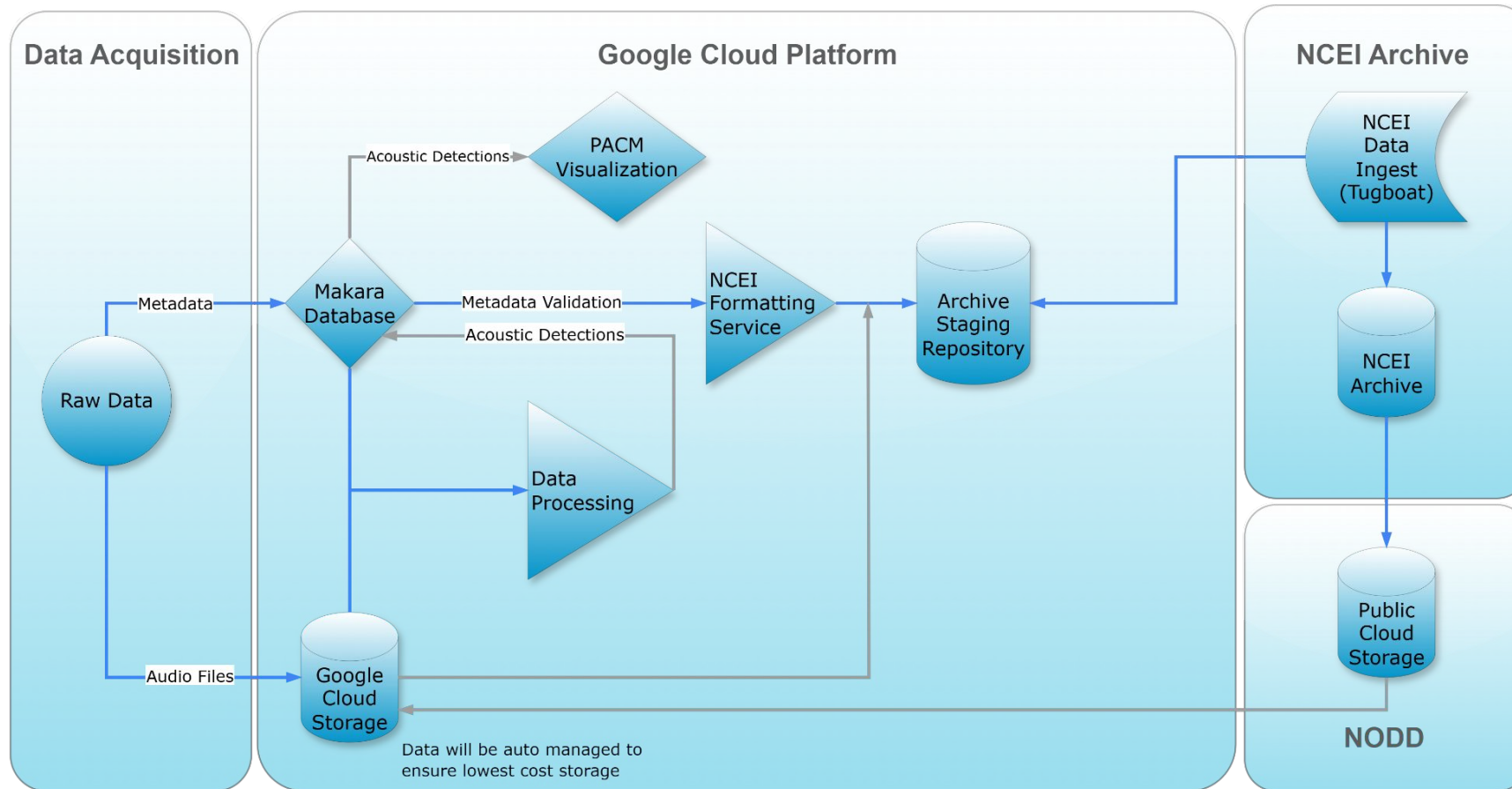
Where the data come from

Where the metadata comes from



How Tugboat Works

PAM-SI Perspective

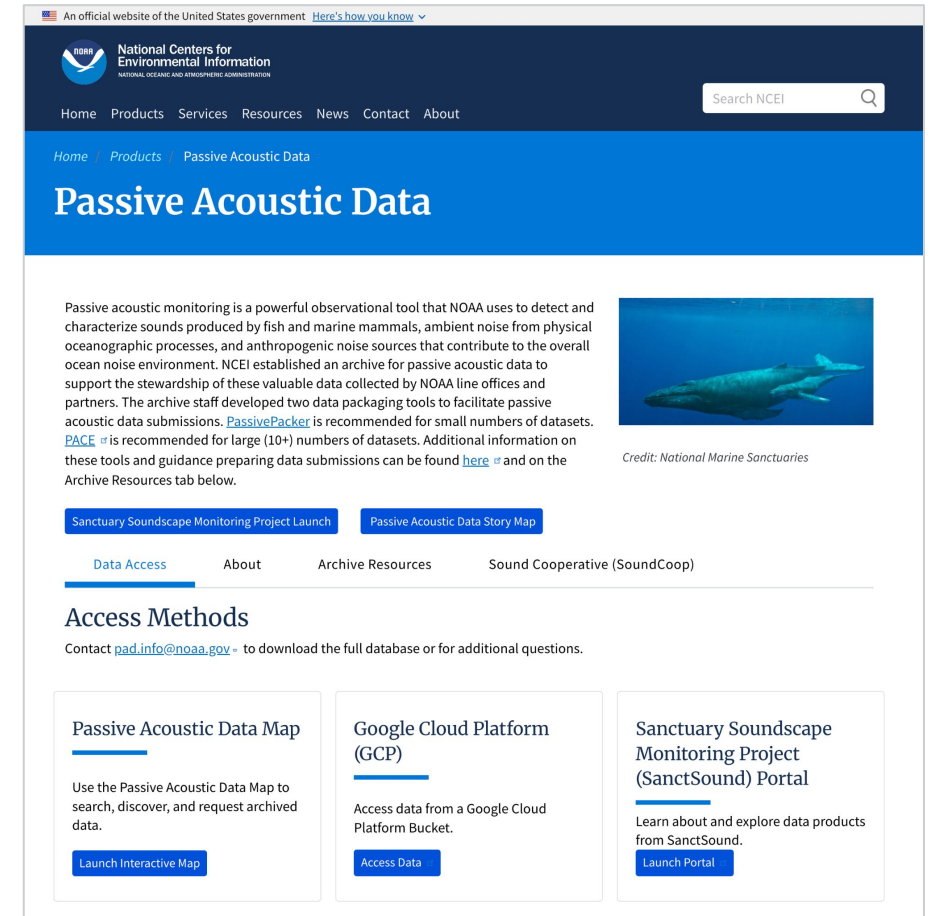


How Tugboat Works

PAM-SI Perspective

NOAA data providers can have their audio files and associated metadata automatically packaged and transmitted to NCEI through the NOAA Fisheries Google Cloud

- Data providers coordinate with NCEI to establish a “Project” within the archive
- Data providers control when the audio data and associated metadata are ready for archiving via the Makara frontend



The screenshot shows the NCEI website's 'Passive Acoustic Data' section. At the top, there's a navigation bar with links for Home, Products, Services, Resources, News, Contact, and About. A search bar is also present. Below the navigation bar, the page title 'Passive Acoustic Data' is displayed. The main content area includes a paragraph explaining passive acoustic monitoring and its use by NOAA. To the right of the text is an image of a whale. Below the text, there are two buttons: 'Sanctuary Soundscape Monitoring Project Launch' and 'Passive Acoustic Data Story Map'. Further down, there's a section titled 'Access Methods' with a contact email 'pad.info@noaa.gov'. At the bottom, there are three boxes: 'Passive Acoustic Data Map' with a 'Launch Interactive Map' button, 'Google Cloud Platform (GCP)' with an 'Access Data' button, and 'Sanctuary Soundscape Monitoring Project (SanctSound) Portal' with a 'Launch Portal' button.

Archiving to NCEI from Makara

Makara frontend website will allow users to submit audio data for archiving to NCEI via Tugboat

- Makara user creates a new “Archive Job”
 - Enter contact info
 - Select NCEI project from dropdown list
 - Select recording entry in Makara
- Metadata then validated against Tugboat requirements, transformed to Passive Acoustic Audio Data submission package, and submitted to Tugboat API
- Makara then tracks the status of each Archive Job, which can be viewed through the frontend

An official website of the United States government [Here's how you know](#)

 **Makara Passive Acoustics Database**
Signed in as [jeffrey.walker@noaa.gov](#)

Home Manage data Query database Lab Manual Templates Guide Contact Us Admin Dashboard

Manage data > Archive to NCEI

Archive to NCEI [New archive job](#)

Package and submit data to the National Centers for Environmental Information (NCEI) archive.

Mockup
These pages show sample data. Nothing is sent to NCEI.

Filter by Status: Select Organization:

ID	Status	Deployment	Recording	Project title	Submitter	Created	Finished
4	PENDING	TEST_GOM_202311_WE03	SOUNDTRAP_RECORDING	NEFSC Passive Acoustic Monitoring of the Northeast U.S. Shelf	Sample Submitter	9/17/2026, 9:40:00 AM	—
3	RUNNING	TEST_GOM_202305_MDR	SOUNDTRAP_RECORDING	NEFSC Passive Acoustic Monitoring of the Northeast U.S. Shelf	Sample Submitter	9/16/2026, 5:05:00 AM	—
2	FAILED	TEST_GOM_202305_MDR	SOUNDTRAP_RECORDING	NEFSC Passive Acoustic Monitoring of the Northeast U.S. Shelf	Sample Submitter	8/28/2026, 2:02:00 PM	8/28/2026, 4:31:00 PM
1	COMPLETED	TEST_GOM_202506_NS01	SOUNDTRAP_RECORDING	NEFSC Passive Acoustic Monitoring of the Northeast U.S. Shelf	Sample Submitter	6/30/2026, 11:44:00 AM	7/2/2026, 7:20:00 AM

Archiving to NCEI from Makara

New Job Submission Form and Validation

New archive job

Submit one recording and its metadata to the NCEI archive. The job runs in the background; follow its progress on the archive job page.

Mockup

This form uses sample projects and recordings. Nothing is sent to NCEI.

Data submitter name *

Jeff Walker

Data submitter email *

jeffrey.walker@noaa.gov

Other people to notify

Optional. Add one email address at a time.

Add

NCEI project title *

Select

NEFSC Passive Acoustic Monitoring of the Northeast X

Organization *

Select the organization that owns the recording you want to archive.

TEST X

Recording *

Select the recording to archive. Each job archives one recording.

TEST_GOM_202305_MDR / SOUNDTRAP_RECORDII X

deployment: TEST_GOM_202305_MDR [Edit deployment](#)

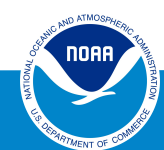
Field	Value	Required by NCEI
deployment_code	TEST_GOM_202305_MDR	Yes
project_code	TEST_GOM	
site_code	MDR	
deployment_platform_type_code	MOORING	Yes
deployment_datetime	2023-05-12T14:00:00Z	Yes
deployment_water_depth_m	92	Yes
deployment_latitude	42.4512	Yes
deployment_longitude	-68.3127	Yes
recovery_datetime	2023-11-03T16:30:00Z	Yes
recovery_latitude	42.4509	Yes
recovery_longitude	-68.3131	Yes

device: ST600_5421 [Edit device](#)

Field	Value	Required by NCEI
device_code	ST600_5421	
device_type_code	SOUNDTRAP	Yes
device_model_number	ST600HF	
device_serial_number	5421	

recording: SOUNDTRAP_RECORDING [Edit recording](#)

Field	Value	Required by NCEI
recording_code	SOUNDTRAP_RECORDING	
recording_start_datetime	2023-05-12T14:00:00Z	Yes
recording_end_datetime	2023-11-03T16:30:00Z	Yes
recording_duration_secs	300	Yes
recording_interval_secs	900	Yes
recording_sample_rate_khz	48	Yes
recording_bit_depth	16	Yes
recording_channel	1	
recording_quality_code	GOOD	
recording_device_depth_m	85	Yes
recording_uri	gs://test-pam/TEST_GOM_202305_MDR/audio/	Yes
recording_calibration_uri	gs://test-pam/TEST_GOM_202305_MDR/calibration/	Yes



Archiving to NCEI from Makara

Tracking Job Status

Archive to NCEI

New archive job

Package and submit data to the National Centers for Environmental Information (NCEI) archive.

Mockup

These pages show sample data. Nothing is sent to NCEI.

Filter by Status

All

Select Organization

TEST

ID	Status	Deployment	Recording	Project title	Submitter	Created	Finished
4	PENDING	TEST_GOM_202311_WE03	SOUNDTRAP_RECORDING	NEFSC Passive Acoustic Monitoring of the Northeast U.S. Shelf	Sample Submitter	9/17/2026, 9:40:00 AM	—
3	RUNNING	TEST_GOM_202305_MDR	SOUNDTRAP_RECORDING	NEFSC Passive Acoustic Monitoring of the Northeast U.S. Shelf	Sample Submitter	9/16/2026, 5:05:00 AM	—
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Archive job 3 RUNNING

Mockup

This page shows sample data. Nothing is sent to NCEI.

This archive job is running. Large recordings can take several days to transfer.

Progress

Step	Status	Finished
1. Validate metadata	DONE	9/16/2026, 5:12:00 AM
2. Build NCEI metadata package	DONE	9/16/2026, 5:12:00 AM
3. Transfer audio files to NCEI	RUNNING	—
4. Wait for NCEI confirmation	WAITING	—

Details

ID
3

Organization
TEST

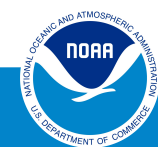
Status
RUNNING

Project title
NEFSC Passive Acoustic Monitoring of the Northeast U.S. Shelf

Deployment
TEST_GOM_202305_MDR

Recording
SOUNDTRAP_RECORDING

Recording ID
1021



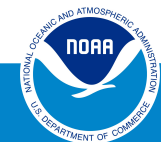
Successes and Next Steps

NCEI

- **Successfully archived** the first passive acoustic and water column sonar datasets submitted via Tugboat in early 2026!
- **October:** Launch User Interface to facilitate metadata management
- **November:** Releasing Email Notification System to provide status updates to users about their submissions
- **December:** Improve the contextual knowledge of the AI metadata validation step
- **2027:** Expanding Tugboat's user base to include OMAO dataset submissions

PAM SI

- Finalizing data transformation pipeline for extracting metadata from Makara and repackaging for Tugboat
- Created UI mockup pages for submitting audio data from Makara to NCEI with automated job tracking
- Last step is to connect new frontend UI to the data transformation pipeline
- Accelerate our ability to make NOAA Fisheries PAM audio data publicly available through NCEI archive



Questions?

Points of contact

Tugboat development

Clint & Quincy: tugboat.info@noaa.gov

Integrating with Tugboat

Hannan: hannan.khan@noaa.gov

Jeff: jeffrey.walker@noaa.gov

